Configure IP Addresses on Mobile Devices

NETWORK 4.2.10

WORK ORDER: CORPORATE IPAD

The employee tablet cannot get a DHCP lease. Configure static IPv4, join the CorpNet Wi-Fi network, and confirm connectivity to the gateway and DNS server.

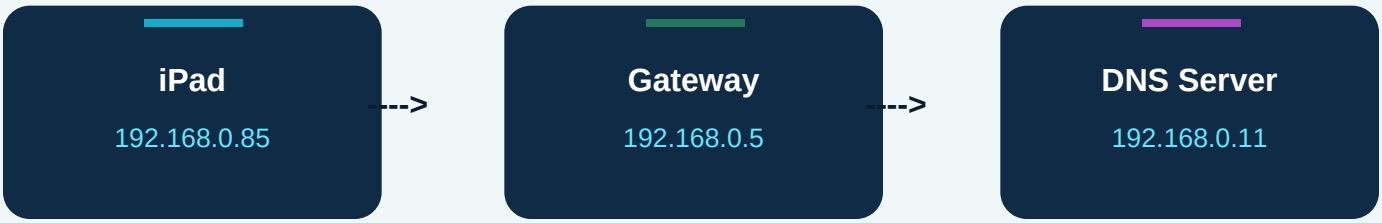
Required static configuration

Wi-Fi SSID: CorpNet
IP address: 192.168.0.85
Subnet mask: 255.255.255.0
Router: 192.168.0.5
DNS server: 192.168.0.11

Learning objectives

Locate the iPad Wi-Fi settings.
Choose Manual instead of Automatic.
Configure static IPv4 and DNS.
Authenticate to CorpNet.
Verify network connectivity.

REFERENCE DIAGRAM 1: IPAD TO CORPORATE NETWORK



The iPad and router are on the same IPv4 /24 network. The router forwards traffic to other networks; the DNS service answers hostname lookups.

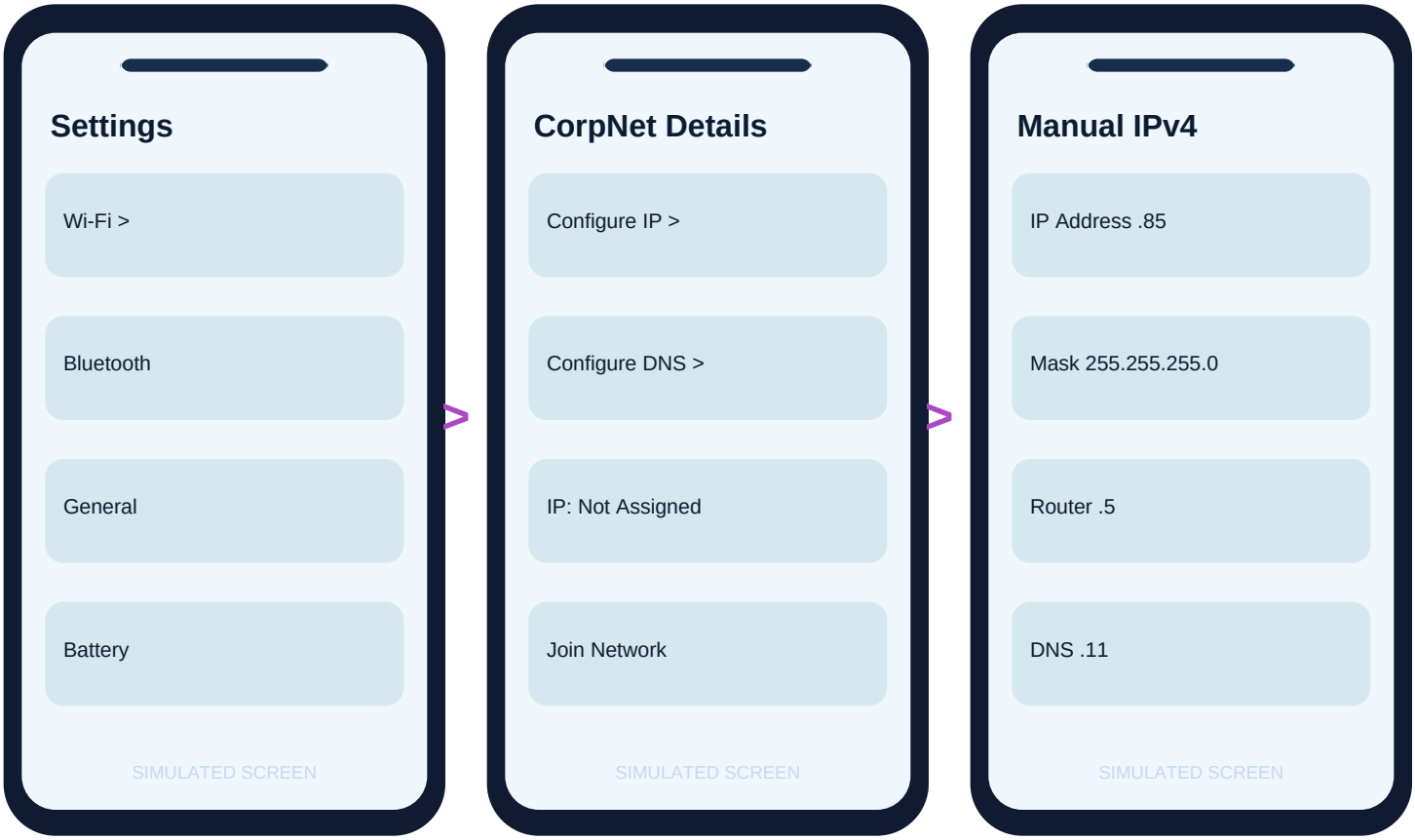
Important concept

A static address must match the correct subnet and must not already be in use. DHCP reservations are usually preferred for ongoing centrally managed deployments.

Finding the iPad Wi-Fi Settings

VISUAL REFERENCE

REFERENCE DIAGRAM 2: SIMULATED IOS NAVIGATION



Configuration path

iPad Home -> Settings -> Wi-Fi -> CorpNet information -> Configure IP -> Manual. Enter the IPv4 address, mask and router. Then choose Configure DNS, enter the DNS server, and join the wireless network.

Screen variation

Actual iPadOS screen names and layout may differ by version. On a real iPad, join and manual addressing steps may appear in a different order.

Network Terms for Mobile Technicians

VOCABULARY

IPv4 address	32-bit address identifying an interface on an IP network.
Subnet mask	Defines which IPv4 bits describe the network and host.
CIDR /24	A mask with 24 network bits, equal to 255.255.255.0.
Default gateway	Router used to reach destinations outside the local subnet.
DNS	Service used to resolve domain names to network addresses.
DHCP	Protocol that normally assigns IP configuration automatically.
Static IP	An administrator-entered address that does not rely on DHCP.
SSID	The name identifying a Wi-Fi network.
Wi-Fi authentication	Verification that a client has permission to join a wireless network.
DHCP lease	Temporary assignment of an address and other network settings.
IP conflict	Two devices attempting to use the same IP address.
Connectivity test	Check of whether a device can communicate with an endpoint.

IP Configuration and Diagnostics

TROUBLESHOOTING

Cannot join CorpNet

Check SSID and Wi-Fi key; join/authentication is distinct from IP configuration.

Connected; cannot reach gateway

Verify 192.168.0.85, /24 mask, and router 192.168.0.5.

Gateway works; names fail

Check DNS 192.168.0.11 and whether that server is reachable.

Wrong subnet

A 10.0.0.x address cannot directly reach 192.168.0.5 without routing.

Duplicate IP warning

Choose a unique unused address or ask IT to reserve the address.

No DHCP lease

Switch to Manual IPv4 using the approved work order values.

NETWORK CHECK ORDER

Wi-Fi association -> IPv4 -> subnet -> gateway -> DNS -> application

Knowledge Check & Student Checklist

REVIEW AND READINESS

1. Why might a technician use static IPv4 instead of DHCP?

2. What subnet prefix matches 255.255.255.0?

3. Which address is the default gateway in this work order?

4. What is the role of the DNS server?

5. Why can a device join Wi-Fi but still fail to reach a server?

Answers for self-check

1. DHCP lease unavailable. 2. /24. 3. 192.168.0.5.
4. Resolves names into addresses. 5. Authentication and IP configuration are separate.

STUDENT READINESS CHECKLIST

[ ] I can locate Configure IP and Configure DNS on an iPad.

[ ] I can explain IP address, subnet mask, router and DNS.

[ ] I can verify an IP configuration and save my MockTIA assessment PDF.